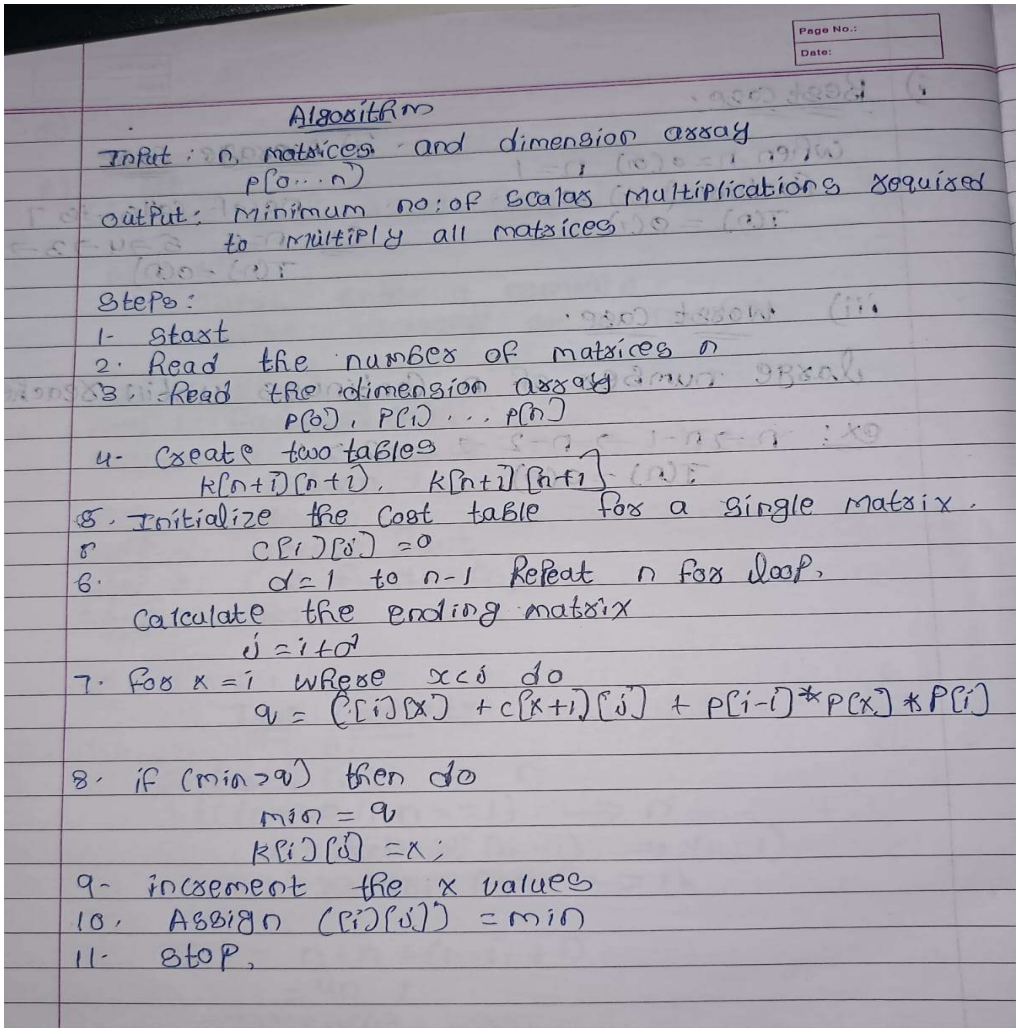
Practical: 6	Implementation of chain matrix multiplication using dynamic programming
01/09/2026	

Aim: To Implement the chain matrix multiplication using dynamic programming

Algorithm:



Algorithm

Input : n, matrices and dimension array p[0...n]

output: Minimum no: of scalas multiplications xequired to multiply all matsices

Steps:

1- Stast
2. Read the numbes of matrices n
3. Read the dimension assay p[0], p[1] ... p[n]
4. Cseate two tables k[n+1][n+1], k[n+1][n+1]
5. Initialize the cost table for a single matsix.
6. c[i][i] = 0
   d = 1 to n-1 Repeat n for loop.
   Calculate the ending matsix
   j = i+d
7. For x = i where x < j do
   q = (c[i][x] + c[x+1][j] + p[i-1] * p[x] * p[j]
8. if (min > q) then do
   min = q
   k[i][j] = x;
9. increment the x values
10. Assign (c[i][j]) = min
11. stop.

Program:

```
#include <iostream>
using namespace std;

int main(){
   int n = 5;

   int p[] = {5, 10, 3, 12, 5, 50};

   int c[10][10] = {0};
   int k[10][10] = {0};

   for(int d = 1; d < n; d++){
for(int i = 1; i <= n-d; i++){
int j = i + d;
         int min = 99999;

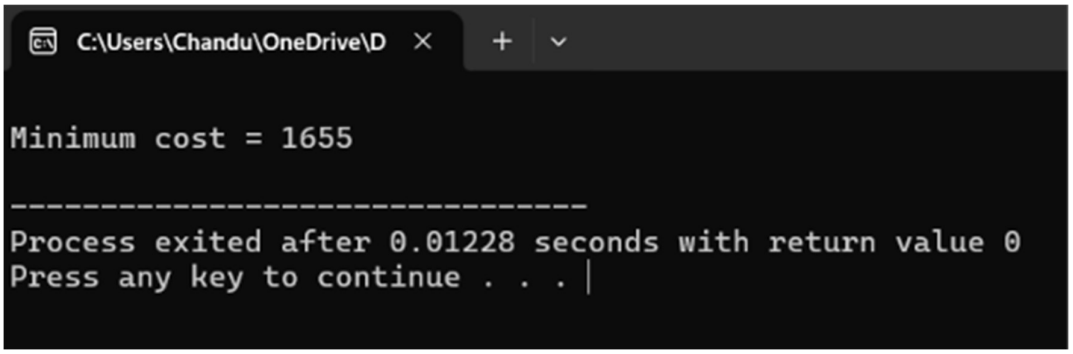
         for(int x = i; x < j; x++){
int q = c[i][x] + c[x+1][j]
               + p[i-1] * p[x] * p[j];

            if(q < min){
min = q;
k[i][j] = x;
            }         }
c[i][j] = min;
      }
   }

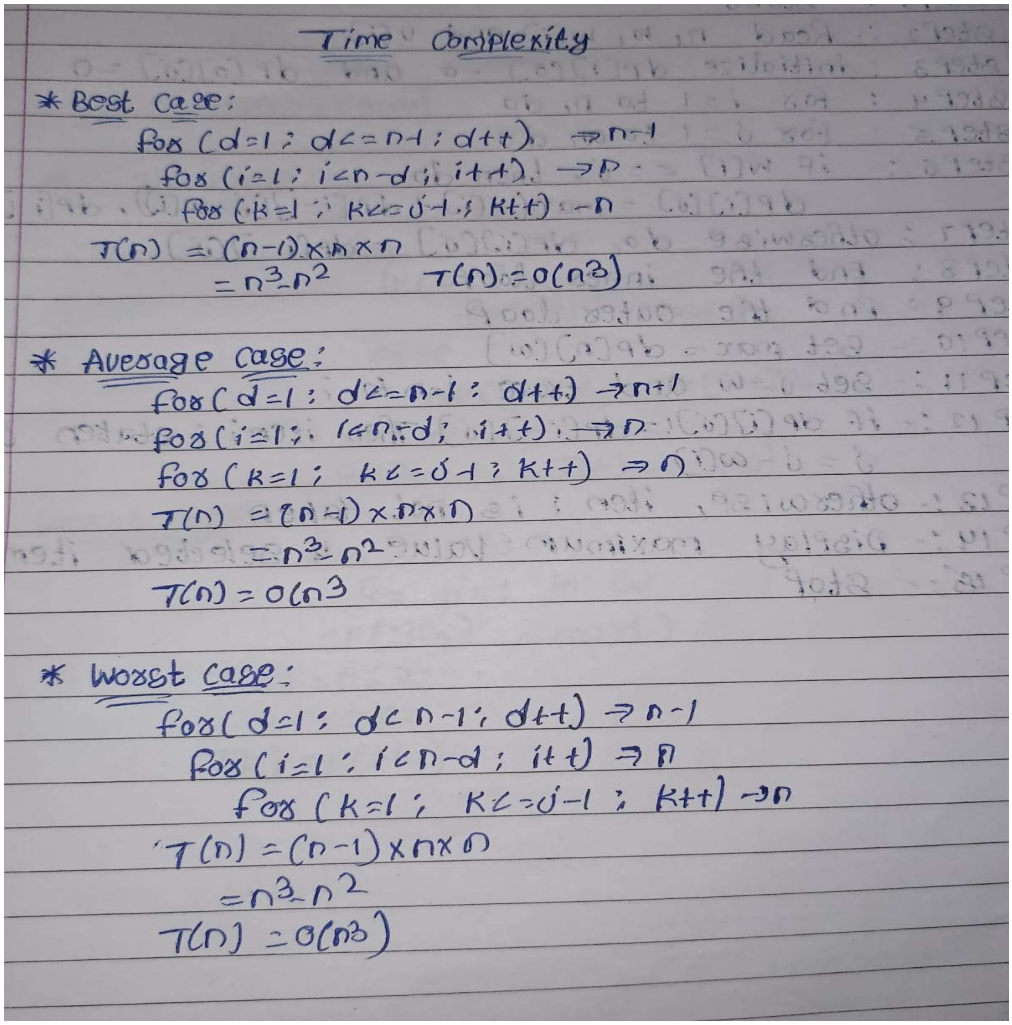
   cout << "\nMinimum cost = " << c[1][n]<<endl;

   return 0;
}
```

Output:



Time Complexity:



Conclusion:

The Implementation of the of chain matrix multiplication using dynamic programming has successfully executed with the time complexity of O(n^3) in the Three cases of best, average and worst. At the end of this program says that the Minimum number of scalar multiplications required to multiply all matrices.